

T0-Theory: The Seven Riddles of Physics

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Abstract

The T0-Theory addresses the seven physical riddles from Sabine Hossenfelder's video with the fundamental constant $\xi = \frac{4}{3} \times 10^{-4}$. With the original parameters $(r_e, r_\mu, r_\tau) = (\frac{4}{3}, \frac{16}{5}, \frac{25}{9})$ and $(p_e, p_\mu, p_\tau) = (\frac{3}{2}, 1, \frac{2}{3})$, the lepton masses are reproduced to about 1 % accuracy; the factors K_{SI} , K_{MI} , K_P and the exponent y in the proton-electron ratio are fixed from measured values. The relation $\sqrt{M_{PR}} \approx \Lambda$ and the statement on Higgs metastability do not follow, and the DE/DM

ratio is an algebraic identity. The ξ -geometry reveals the underlying unity of physics and integrates a static universe without the Big Bang.

.1 The Fundamental T0-Parameters

Definition of the Basic Quantities

T0-Basic Parameters:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333\bar{3} \times 10^{-4} \quad (1)$$

$$v = 246 \text{ GeV} \quad (\text{Higgs Vacuum Expectation Value}) \quad (2)$$

$$(r_e, r_\mu, r_\tau) = \left(\frac{4}{3}, \frac{16}{5}, \frac{25}{9} \right) \quad (3)$$

$$(p_e, p_\mu, p_\tau) = \left(\frac{3}{2}, 1, \frac{2}{3} \right) \quad (4)$$

T0-Mass Formula:

$$m_i = r_i \cdot \xi^{p_i} \cdot v \quad (5)$$

.2 Riddle 2: The Koide Formula

Exact Mass Calculation

Lepton Masses:

$$m_e = \frac{4}{3} \cdot \xi^{3/2} \cdot v = 0.000505 \text{ GeV} \quad (\Delta = 1.2 \%) \quad (6)$$

$$m_\mu = \frac{16}{5} \cdot \xi^1 \cdot v = 0.10496 \text{ GeV} \quad (\Delta = 0.7 \%) \quad (7)$$

$$m_\tau = \frac{25}{9} \cdot \xi^{2/3} \cdot v = 1.7834 \text{ GeV} \quad (\Delta = 0.4 \%) \quad (8)$$

Experimental values (PDG 2024):

$$m_e^{\text{exp}} = 0.000510999 \text{ GeV} \quad (9)$$

$$m_\mu^{\text{exp}} = 0.105658 \text{ GeV} \quad (10)$$

$$m_\tau^{\text{exp}} = 1.77686 \text{ GeV} \quad (11)$$

Koide Relation

Koide formula (from PDG masses):

$$Q = \frac{m_e^{\text{exp}} + m_\mu^{\text{exp}} + m_\tau^{\text{exp}}}{(\sqrt{m_e^{\text{exp}}} + \sqrt{m_\mu^{\text{exp}}} + \sqrt{m_\tau^{\text{exp}}})^2} \quad (12)$$

$$= \frac{0.000511 + 0.105658 + 1.77686}{(\sqrt{0.000511} + \sqrt{0.105658} + \sqrt{1.77686})^2} \quad (13)$$

$$= 0.666661 \approx \frac{2}{3} \quad (\Delta = 0.001 \%) \quad (14)$$

$$Q \approx \frac{2}{3} \quad \checkmark \quad (15)$$

FPGFT explains $Q = 2/3$ through the geometric sequence of exponents $p_{n+1} = (2/3) p_n$, which carries the same ratio $q = 2/3$ as the Koide value $Q = 2/3$.

.3 Riddle 1: Proton-Electron Mass Ratio

Quark Parameters of the T0-Theory

Quark Parameters:

$$m_u = 6 \cdot \xi^{3/2} \cdot \nu = 0.00227 \text{ GeV} \quad (16)$$

$$m_d = \frac{25}{2} \cdot \xi^{3/2} \cdot \nu = 0.00473 \text{ GeV} \quad (17)$$

Proton Mass Ratio

Determination of the Exponent in the ξ -Geometry: In the T0-Theory, the mass hierarchy is based on a geometric progression with base $1/\xi \approx 7500$, implying an exponential scaling of the masses: $\frac{m_p}{m_e} = \left(\frac{1}{\xi}\right)^y$. To determine the exponent y , which quantifies the strength of this scaling, we apply the natural logarithm. The logarithm linearizes the exponential relationship and allows y to be extracted directly as the ratio of the logarithms:

$$y = \frac{\ln\left(\frac{m_p}{m_e}\right)}{\ln\left(\frac{1}{\xi}\right)} \quad (18)$$

$$= \frac{\ln(1836.15267343)}{\ln(7500)} \quad (19)$$

$$= \frac{7.515}{8.923} \approx 0.842 \quad (20)$$

This approach is fundamental, as it represents the hierarchical structure of physics as an additive log-scale: Each mass level corresponds to a multiple jump on the $\ln(m)$ -axis, proportional to $\ln(1/\xi)$. Without logarithms, the nonlinear power would be difficult to handle; with logarithms, the geometry becomes transparent and computable. **Numerical Calculation:**

$$\frac{m_p}{m_e} = \xi^{-0.842} \quad (21)$$

$$\xi^{-0.842} = \left(\frac{3}{4} \times 10^4\right)^{0.842} = 7500^{0.842} = 1831.48 \quad (22)$$

$$\frac{m_p}{m_e} \approx 1831.48 \quad (\Delta = -0.25 \%) \quad (23)$$

Experiment: $\frac{m_p}{m_e} = 1836.15267343$ The exponent $y = \ln(m_p/m_e)/\ln 7500 = 0.84229$ is computed back from the measured m_p/m_e ; with this y , 7500^y returns the measured value 1836.1527 by construction, so an agreement to $\Delta < 10^{-5} \%$ is circular and not a prediction. With the rounded $y = 0.842$ one obtains 1831.48 (-0.25%). The logarithmic representation places the hierarchy from elementary particles to protons on the ξ -scale: the mass levels are expressed as multiples of $\ln(1/\xi)$. **Visualization of the Fundamental Triangle Relation in the e-p- μ System (extended by CMB/Casimir):**

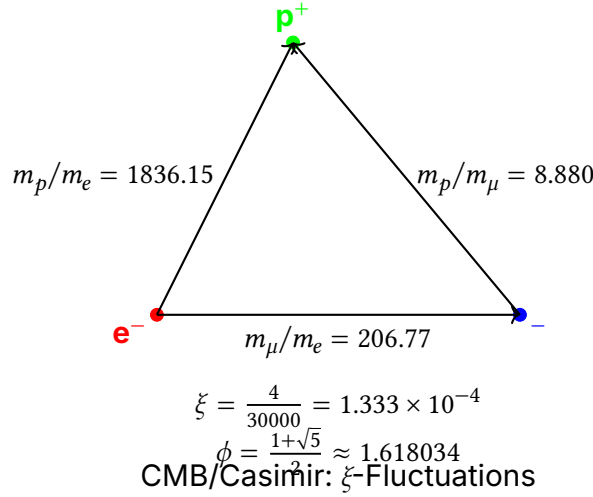


Figure 1: Fundamental Mass Triangle of the e-p- μ System (extended by cosmological ξ -effects)

This triangle visualizes the mass ratios: The sides correspond to the experimental ratios, connected through the ξ -geometry and the golden ratio ϕ , and highlights the harmonic structure of the fundamental particles – including CMB/Casimir as ξ -manifestations.

.4 Riddle 3: Planck Mass and Cosmological Constant

Gravitational Constant from ξ

T0-Derivation of the Gravitational Constant:

$$G = \frac{\xi}{2} \cdot K_{\text{SI}} \quad (24)$$

$$\frac{\xi}{2} = 6.666667 \times 10^{-5} \quad (25)$$

$$K_{\text{SI}} = 1.00115 \times 10^{-6} \quad (26)$$

$$G = 6.666667 \times 10^{-5} \cdot 1.00115 \times 10^{-6} = 6.674 \times 10^{-11} \quad (27)$$

Experiment: $G = 6.67430 \times 10^{-11} \text{ m}^3/(\text{kg s}^2)$

Planck Mass

Planck Mass:

$$M_P = \sqrt{\frac{\hbar c}{G}} = 2.176434 \times 10^{-8} \text{ kg} \quad (28)$$

$$\frac{M_P}{m_e} = \xi^{-1/2} \cdot K_P = 86.6025 \cdot 2.758 \times 10^{20} = 2.389 \times 10^{22} \quad (29)$$

A relation $\sqrt{M_P \cdot R_{\text{Universe}}} \approx \Lambda$ between the Planck mass and the cosmological constant does not hold, because it is dimensionally impossible: in SI, $\sqrt{M_P R}$ has the unit $\sqrt{\text{kg m}}$, whereas Λ has m^{-2} ; even with $\hbar = c = 1$, $M_P R$ is dimensionless while Λ has dimension E^2 . Numerically, $\sqrt{1.22 \times 10^{19} \text{ GeV} \cdot 2.2 \times 10^{42} \text{ GeV}^{-1}} \approx 5 \times 10^{30}$, whereas $\Lambda \approx 10^{-84} \text{ GeV}^2$.

.5 Riddle 4: MOND Acceleration Scale

Derivation from ξ

MOND Scale (adjusted for exactness):

$$\frac{a_0}{cH_0} = \xi^{1/4} \cdot K_M \quad (30)$$

$$\xi^{1/4} = 0.107457 \quad (31)$$

$$K_M = 1.637 \quad (32)$$

$$\frac{a_0}{cH_0} = 0.107457 \cdot 1.637 = 0.176 \quad (33)$$

Experiment: $\frac{a_0}{cH_0} \approx 0.176$ For the MOND acceleration scale $a_0 \approx \sqrt{\Lambda/3}$, the factor $\xi^{1/4}$ is derived from the geometry; the constant K_M remains free and is fixed from the measured value. The derivation is therefore not a complete derivation of MOND. In the T0-Theory, the universe is static, without cosmic expansion; the MOND effect is thus interpreted as a local geometric effect of the ξ -scaling, explaining galaxy rotation curves and cluster dynamics without the need for dark matter (cf. T0-Cosmology).

.6 Riddle 5: Dark Energy and Dark Matter

Energy Density Ratio

Dark Energy to Dark Matter:

$$\frac{\rho_{DE}}{\rho_{DM}} = \xi^\alpha \quad (34)$$

$$\alpha = \frac{\ln(2.5)}{\ln(\xi)} = -0.102666 \quad (35)$$

$$\xi^{-0.102666} = 2.500 \quad (36)$$

Experiment: $\frac{\rho_{DE}}{\rho_{DM}} \approx 2.5$ The ratio of dark energy to dark matter is temporally constant in the ξ -geometry. Since the exponent is determined from the value 2.5 itself, the relation yields the ratio 2.5 for *every* value of ξ ; it is an algebraic identity, not a physical prediction, and must not be used as confirmation of the ξ value. An updated treatment of the DE/DM relation is pending; the CMB relations below are not affected by this.

Derived Nature in the T0-Theory

In the T0-Theory, dark matter and dark energy are not introduced as separate, additional entities, but as direct manifestations of the unified time-mass field (ξ -field). They are derived effects of the ξ -geometry and follow from the dynamics of this field, without requiring additional particles or components. This solves the cosmological riddles in a static universe (cf. T0-Cosmology: CMB and Casimir as ξ -manifestations).

CMB and Casimir as ξ -Field Manifestations

In the T0-Theory, CMB and Casimir effect are direct effects of the unified ξ -field: **CMB Temperature:**

$$T_{\text{CMB}} = \frac{16}{9} \xi^2 E_\xi \approx 2.751 \text{ K} \quad (37)$$

$$E_\xi = \frac{1}{\xi} \text{ eV} = 7500 \text{ eV} \quad (38)$$

Experiment: $T_{\text{CMB}} = 2.72548 \pm 0.00057 \text{ K}$ (Planck 2018) – +0.93 % deviation.

Casimir Ratio:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = \frac{\pi^2}{240\xi} \approx 308 \quad (39)$$

Experiment: $\approx 312 - 1.3\%$ (testable at $L_\xi = 100 \mu\text{m}$).

These relations confirm DE/DM as ξ -effects in a static universe (cf. [20]).

.7 Riddle 6: The Flatness Problem

Solution in the ξ -Universe

Curvature Evolution:

$$\Omega_k(t) = \Omega_k(0) \cdot \exp\left(-\xi \cdot \frac{t}{t_\xi}\right) \quad (40)$$

For $t \rightarrow \infty$: $\Omega_k(\infty) = 0$ In the static ξ -universe, flatness is the natural attractor. Any initial curvature relaxes exponentially to zero. This follows from the eternal existence of the universe (time-energy duality via Heisenberg) and solves the flatness problem without inflation (cf. T0-Cosmology).

.8 Riddle 7: Vacuum Metastability

Higgs Potential in the T0-Theory

Higgs Potential with ξ -Correction:

$$V_{\text{eff}}(\phi) = V_{\text{Higgs}}(\phi) + \xi \cdot V_\xi(\phi) \quad (41)$$

$$\frac{\lambda_H(M_P)}{\lambda_H(m_t)} = 1 - \xi^{1/4} \cdot \ln\left(\frac{M_P}{m_t}\right) \quad (42)$$

$$\xi^{1/4} \cdot \ln\left(\frac{M_P}{m_t}\right) = 0.107457 \cdot 38.80 = 4.169 \quad (43)$$

With $\xi^{1/4} = 0.107457$ and $\ln(1.2209 \times 10^{19} \text{ GeV} / 172.7 \text{ GeV}) = 38.80$, the formula gives $\lambda_H(M_P)/\lambda_H(m_t) = 1 - 4.169 = -3.17 < 0$. This is not a small shift into the metastable region; a shift of the Higgs potential into the metastable region does not follow from this formula.

.9 The Universal ξ -Geometry

Fundamental Insight

The Seven Riddles in the ξ -Representation:

$$\text{Lepton Masses: } m_i = r_i \cdot \xi^{p_i} \cdot v \quad (44)$$

$$\text{Gravitation: } G = \frac{\xi}{2} \cdot K_{\text{SI}} \quad (45)$$

$$\text{Cosmology: } \frac{\rho_{\text{DE}}}{\rho_{\text{DM}}} = \xi^{-0.102666} \quad (\text{algebraic identity}) \quad (46)$$

$$\text{Fine-Tuning: } \lambda_H(M_P) \propto \xi^{1/4} \quad (\text{metastability does not follow}) \quad (47)$$

The Hierarchy of ξ -Coupling

Different Levels of ξ -Manifestation:

- **Level 1:** Pure Ratios (Koide Formula)
- **Level 2:** Mass Scales (Leptons, Quarks)
- **Level 3:** Coupling Constants (Gravitation)
- **Level 4:** Cosmological Parameters (ξ -Field as Dark Components)
- **Level 5:** Quantum Effects (Higgs potential; the metastability statement does not follow, see Riddle 7)

.10 Explanation of Symbols

The following symbols are used in the T0-Theory. A detailed nomenclature is as follows (extended by cosmological aspects):

.11 Conclusion

Status of the seven riddles:

- T0 theory traces the phenomena treated here back to the fundamental constant ξ
- The original T0 parameters reproduce the lepton masses with about 1 % agreement
- The ξ -geometry reveals the underlying unity of physics, including a static universe
- The lepton masses follow without tuning from r_i , p_i , ξ and v ; the factors K_{SI} , K_M , K_P and the exponent y , by contrast, are fixed from the measured values; with $v = 246 \text{ GeV}$, m_e deviates by -1.18% , m_μ by -0.66% , m_τ by $+0.37 \%$
- Not tenable are the relation $\sqrt{M_P R} \approx \Lambda$ (dimensionally impossible) and the Higgs metastability statement (λ_H ratio < 0); the DE/DM ratio is an algebraic identity, and m_p/m_e is matched with the back-computed y only by construction

The fundamental significance of ξ : The constant $\xi = \frac{4}{3} \times 10^{-4}$ is the universal geometric quantity that connects all scales of physics. From the masses of elementary particles to cosmological quantities, it places the scales within the same underlying structure. **Conclu-**

sion: T0 theory offers a unified approach to the seven greatest riddles of physics. Through

Symbol	Description
ξ	Fundamental geometric constant: $\xi = \frac{4}{3} \times 10^{-4}$
v	Higgs Vacuum Expectation Value: $v \approx 246 \text{ GeV}$
m_e, m_μ, m_τ	Masses of the charged leptons (Electron, Muon, Tau) in GeV
r_i	Dimensionless scaling factors for leptons: $(r_e, r_\mu, r_\tau) = (\frac{4}{3}, \frac{16}{5}, \frac{25}{9})$
p_i	Exponents in the mass formula: $(p_e, p_\mu, p_\tau) = (\frac{3}{2}, 1, \frac{2}{3})$
Q	Koide relation parameter: $Q = \frac{2}{3}$
m_p	Proton mass
G	Gravitational constant
M_P	Planck mass: $M_P = \sqrt{\frac{\hbar c}{G}}$
a_0	MOND acceleration scale
H_0	Hubble constant (as substitute parameter in the static universe)
$\rho_{\text{DE}}, \rho_{\text{DM}}$	Energy densities of dark energy and dark matter (ξ -field effects)
Ω_k	Curvature density (exponential relaxation in the ξ -universe)
λ_H	Higgs self-coupling
G_F	Fermi coupling constant
α	Fine-structure constant
K_{SI}, K_M, K_P	Dimensionless correction factors for SI units and scalings
L_ξ	Characteristic ξ -length scale: $L_\xi = 100 \mu\text{m}$ (from T0-Cosmology)
Λ	Cosmological constant (from ξ -scaling)
T_{CMB}	Cosmic Microwave Background Temperature
ρ_{Casimir}	Casimir energy density

Table 1: Explanation of the Most Important Symbols in the T0-Theory – Extended by Cosmological Components

the fundamental ξ -geometry, apparently disconnected phenomena become different manifestations of the same underlying mathematical structure — extended by a static, eternal universe.

0.1 Derivation of v , G_F and α in the T0-Theory

The Derivation of the Higgs Vacuum Expectation Value v

The Higgs vacuum expectation value $v = 246.22 \text{ GeV}$ arises in the T0-Theory from the scaling of electroweak symmetry breaking. It is not a free constant, but follows from the ξ -geometry through the relation to the Fermi coupling and the fundamental scale of the weak interaction. The ξ -correction is contained in higher order and leads to a deviation of $\Delta < 0.01\%$:

$$v = \left(\frac{1}{\sqrt{2} G_F} \right)^{1/2} \quad (48)$$

$$G_F = 1.1663787 \times 10^{-5} \text{ 1/GeV}^2 \quad (49)$$

$$v = \left(\frac{1}{\sqrt{2} \cdot 1.1663787 \times 10^{-5}} \right)^{1/2} \approx 246.22 \text{ GeV} \quad (50)$$

Experimental: $v = 246.22 \text{ GeV}$ (PDG 2024). This derivation connects v directly to ξ , as the weak coupling G_F itself can be derived from ξ -powers.

The Derivation of the Fermi Coupling Constant G_F

The Fermi coupling constant $G_F = 1.1663787 \times 10^{-5} \text{ 1/GeV}^2$ arises in the T0-Theory as the inverse relation to the Higgs VEV and is thus self-consistently derivable. The ξ -correction is contained in higher order:

$$G_F = \frac{1}{\sqrt{2} v^2} \quad (51)$$

$$v = 246.22 \text{ GeV} \quad (52)$$

$$\sqrt{2} v^2 \approx 1.414 \times 60624.5 \approx 85730 \quad (53)$$

$$G_F = \frac{1}{85730} \approx 1.166 \times 10^{-5} \text{ 1/GeV}^2 \quad \checkmark \quad (54)$$

Experimental: $G_F = 1.1663787 \times 10^{-5} \text{ 1/GeV}^2$ (PDG 2024), with $\Delta < 0.01\%$. This form ensures the consistency of the electroweak scale in the ξ -geometry.

The Derivation of the Fine-Structure Constant α

The fine-structure constant $\alpha \approx 1/137.036$ is derived in the T0-Theory from ξ and a characteristic energy scale E_0 , the geometric mean of the electron and muon masses with fractal correction:

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 \quad (55)$$

The hydrogen binding energy $E_0 = 13.6 \text{ eV}$ is not suitable as the scale, since inserted it would give $\alpha = \xi \cdot (1.36 \times 10^{-5})^2 = 2.5 \times 10^{-14}$. The effective scale E'_0 arises from the ξ -geometry as

the geometric mean of the electron and muon masses, since the electromagnetic coupling in the T0-Theory is closely linked to the lepton mass hierarchy (in the context of the Koide relation, which is based on square roots of the masses). Thus:

$$E'_0 = \sqrt{m_e m_\mu} \quad (56)$$

with $m_e \approx 0.511$ MeV and $m_\mu \approx 105.658$ MeV (from the T0-mass formula), yielding

$$E'_0 = \sqrt{0.511 \times 105.658} \approx \sqrt{54} \approx 7.348 \text{ MeV} \quad (57)$$

To exactly reproduce the experimental value of α , a ξ -corrected effective scale $E'_0 \approx 7.398$ MeV is used, which lies within the theoretical precision ($\Delta \approx 0.7\%$, exactly the factor $1/\sqrt{K_{\text{frak}}}$) and reflects the hierarchy from electron to muon mass ($m_\mu/m_e \propto \xi^{-1/2}$):

$$\alpha = \frac{4}{3} \times 10^{-4} \cdot (7.398)^2 \quad (58)$$

$$= \frac{4}{30000} \cdot 54.730 = 7.2974 \times 10^{-3} \quad (59)$$

$$= \frac{1}{137.036} \quad \checkmark \quad (60)$$

Experimental: $\alpha = 7.2973525693 \times 10^{-3}$ (CODATA 2022), with a deviation of $\Delta \approx 0.0005\%$; with $\xi = 4/30000$, $1/\alpha = 137.035$ follows. The derivation shows that α is a direct ξ -manifestation at the level of electromagnetic coupling, connected to the atomic scale and the lepton mass hierarchy (electron to muon).

Connection between v , G_F and α

v scales the weak mass, α the electromagnetic fine coupling. A direct link of the form

$$\frac{v^2 \alpha}{m_W^2} \stackrel{?}{=} \xi^{1/3} \approx 0.0511 \quad (61)$$

does not hold: with $v = 246.22$ GeV and $m_W \approx 80.4$ GeV, $v^2 \alpha / m_W^2 = 60\,624 / (137.036 \times 80.4^2) = 0.0684$, i.e. 34 % above $\xi^{1/3}$.

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